Research Statement

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Research interests

My research lies at the interface of holography, quantum information, quantum field theory, and mathematical physics. I am interested in the basic question: *Which structures in the boundary quantum theory encode the dynamics and internal geometry of black holes?* My approach is to start from quantities that are intrinsically defined in the quantum theory: thermal partition functions, spectral measures, Krylov dynamics, and generalized entropy monotones; and ask what gravitational information can be reconstructed from them. This perspective has led me from perturbative quantum field theory to holographic complexity, black-hole thermodynamics, as well as to spectral and orthogonal-polynomial methods.

Krylov complexity from partition functions and bulk geometry

My main recent work with Aranya Bhattacharya, Mario Flory, Michal P. Heller, and Tim Schuhmann [1] develops a systematic reconstruction procedure between boundary complexity and bulk geometry. In particular, we provided a critical analysis of a partition-function construction of Krylov spread complexity for thermofield-double states that provides a dimension-independent boundary reconstruction from semiclassical holographic partition functions, while developing the bulk construction for the BTZ saddle. Using double-scaled SYK as a benchmark, we showed that taking the classical limit too early, i.e., at the level of individual Lanczos coefficients, can discard information that contributes at leading order once the full complexity is assembled. We then applied the reconstruction to a large-central-charge CFT_2 state dual to the BTZ black hole. The resulting high-order early-time expansion and its continuation indicate a thermal-scale crossover compatible with a return toward quadratic pre-saturation growth, in contrast with the generic linear late-time behavior of maximal volume and finite-function Complexity=Anything observables. We then found that an infinite tower of extrinsic-curvature invariants, fixed order by order by the boundary series, can produce a resummed observable with the singular late-time structure required for quadratic behavior. I regard this as an existence result rather than a unique holographic dictionary entry. More importantly, it turns the problem into a sharper question: What principles select the admissible bulk representation of a boundary quantity that is already defined independently of gravity?

Broader foundations and current directions

My earlier work with Dario Zappalà on Lorentz-symmetry-violating Lifshitz-type field theories [2] gave me a complementary foundation in perturbative quantum field theory. We studied $3+1$-dimensional anisotropic theories with dynamical exponent $z = 3$, where higher spatial derivatives soften ultraviolet divergences without introducing higher time derivatives. I worked with loop power counting and the renormalization of scalar and Yukawa-like interactions, including the conditions under which the corresponding couplings become asymptotically free.

More recently, with Alice Bernamonti and Federico Galli, I have been extending information-theoretic constraints on black-hole dynamics to charged AdS black holes [4], building on the Rényi second-law framework of Ref. [3]. Our analysis uses the full family of Rényi entropies, together with the Hawking–Page structure of the replicated geometry, to constrain mergers with equal, unequal, and opposite charges and to study extremal limits and bounds on the final state. This line broadens my use of quantum information

beyond complexity, using generalized entropic quantities to constrain which gravitational processes are compatible with information-theoretic monotonicity and phase structure.

At the same time, I am developing a more analytical formulation of the partition-function programme itself. Rather than producing a long finite Lanczos chain and only then studying its asymptotics, I want to work directly with the spectral measure to explore the large-scale structure of Krylov dynamics and to separate universal spectral information from artifacts of finite-order reconstruction. This is the bridge between my holographic work and the spectral/statistical-physics direction of my future research.

Future research: motivation and programme

My postdoctoral programme is organized around two related questions. First, *how much of black-hole dynamics can be reconstructed from thermal and spectral data without guessing the bulk observable in advance?* Second, *which features of Krylov transport are genuinely holographic, and which belong to broader universality classes of spectral dynamics?* I see these questions as complementary rather than as two separate projects: the spectral analysis should constrain what a bulk dual is allowed to do, while holographic examples provide sharp tests of which spectral structures have geometric meaning.

Holography and black-hole dynamics

My first postdoctoral objective would be to determine how universal the partition-function reconstruction of Krylov dynamics is across holographic theories. The BTZ case is a controlled starting point, but a convincing holographic dictionary should survive changes of state and dimension. I plan to study higher-dimensional AdS black holes and more general three-dimensional configurations, including rotating or boosted BTZ geometries and perturbations of thermofield-double states.

A second direction concerns the structure of the bulk observable itself. In higher dimensions there are many more intrinsic and extrinsic curvature invariants, leading to greater freedom for the construction of a generalized observable. A key question is whether the asymptotics of the boundary reconstruction determine universal properties of the late-time accumulation surface that can constrain this freedom, and whether the singular resummation found in BTZ has a geometric counterpart that persists in more realistic black-hole backgrounds.

A third goal is to go beyond the leading semiclassical saddle. Because the boundary construction takes the partition function as its input, subleading saddle contributions and finite-c corrections should, in principle, induce definite corrections to the Krylov dynamics. This offers a concrete way to ask how a semiclassical bulk description deforms as genuinely quantum information enters the partition function.

Spectral and statistical-physics foundations

A natural mathematical-physics direction of my work is to formulate Krylov dynamics directly as a problem in spectral analysis. The Lanczos recursion is the three-term recurrence relation of orthogonal polynomials associated with the spectral measure of the chosen state. This makes complexity growth accessible through recurrence coefficients, spectral edges, and asymptotic measures rather than through brute-force time evolution. I plan to develop the Edgeworth and large-deviation approaches into systematic tools for deriving both fixed-index and continuum Lanczos behavior directly from partition functions. In particular, Edgeworth expansions organize deviations from Gaussianity and give controlled access to low-order recurrence data, whereas a large-deviation potential can retain hard edges and other features that are invisible at every finite order in the perturbative expansion. Understanding where these descriptions

overlap, and where they differ, may provide a general way to infer complexity growth from limited thermodynamic information in strongly coupled systems.

I am also developing a related project on connections among Krylov complexity, orthogonal polynomials, and the asymmetric simple exclusion process (ASEP), motivated by the appearance of structures related to JT/KPZ physics. Askey–Wilson polynomials, edge universality, and Tracy–Widom-type fluctuations suggest that tools from integrable stochastic systems may classify distinct regimes of Krylov transport. The aim would be to identify which spectral mechanisms behind complexity growth are universal across holographic, many-body, and stochastic settings. This direction would let me combine exact solvable models with asymptotic analysis in regimes where direct diagonalization is inaccessible.

Selected references. [1] A. Bhattacharya, M. Flory, M. P. Heller, E. Rizza, T. Schuhmann, arXiv:2608.09922 (2026). [2] E. Rizza, D. Zappalà, *Mod. Phys. Lett. A* **37**, 2250203 (2022), arXiv:2209.11060. [3] A. Bernamonti, F. Galli, R. C. Myers, I. A. Reyes, *JHEP* **10** (2024) 177, arXiv:2407.01753. [4] A. Bernamonti, F. Galli, E. Rizza, *Charged AdS Black Hole*, work in progress (2026).